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COLLEGE OF ARTS AND SCIENCES

FOOD SECURITY AND POVERTY: ASSESSING HOUSEHOLD EXPERIENCES
OF FOOD SHORTAGE

A Final Requirement for the Survey Operation Course
in the Bachelor of Science in Statistics degree
Presented to the College of Arts and Sciences

Submitted to:

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	Introduction Food security is a fundamental requirement for every household, yet food shortages remain a common issue in many communities. Food insecurity occurs when families lack stable and sufficient access to safe and nutritious food. This condition affects daily living, work performance, and overall well-being. Because of this, it is important to understand the factors that contribute to food shortages and how these factors differ among households. In the context of poverty, food access becomes more limited, making it relevant to study how household characteristics influence food availability. One of the major factors often linked to food insecurity is monthly household income. However, income alone may not fully explain why some families experience food shortages while others do not. Some households may have larger family sizes, multiple dependents, or only one income earner supporting the entire household. Others may spend a significant portion of their budget on food but still struggle to meet their needs. These conditions show that food insecurity is a result of different factors that may interact with one another. Understanding these relationships is necessary to give a clearer assessment of why food shortage occurs (Capistrano, 2021; Lim, 2014). This study aims to find out if monthly household income is connected to food insecurity (Akinbamini et al., 2025). It also considers household size, number of	

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	earners, and food expenses. The study asks if low income directly causes food shortages, or if other household traits matter more. By looking at these factors together, the study will see which ones relate most to food insecurity. This helps show if food shortage is mainly an income issue or if other factors play a big role. There is a research gap because little data connects monthly income and food insecurity in Mandaluyong. No published studies look at the combined effect of household size, number of earners, and food expenses in this area. Current literature also lacks local data for planning poverty and food security programs. This gap means a study with area-specific findings is needed. Filling this gap will give information for future decisions and community strategies. Overall, this study focuses on the problem of food shortage. By analyzing the socioeconomic factors of households (Bronson & Knezevic, 2016), the research aims to explain why some families experience food insecurity and how various household characteristics influence this condition. Statement of the Problem This study aims to determine whether monthly household income is associated with the level of household food insecurity, as measured by the Household Food Insecurity Access Scale (HFIAS) score. Although income is known to influence access to adequate food, this relationship may vary depending on other household characteristics. Factors such as household size, number of	

income earners, and monthly food expenditure may also affect the degree of food insecurity and, therefore, need to be statistically controlled.

This study seeks to answer the following question:

1. What is the Socioeconomic profile of Households in Mandaluyong City in terms of:
 - 1.1 Estimated Monthly Household Income;
 - 1.2 Household Size;
 - 1.3 Number of Income Earners; and
 - 1.4 Monthly Household Food Expenditure
2. Is there a significant difference in HFIAS scores across households with different levels of monthly household income?
3. Do the covariates: Household size, Number of Income Earners, and Monthly Household Food Expenditure significantly predict or are significantly associated with Household Food Insecurity (HFIAS scores)?

Hypotheses

H₀₁: There is no significant difference in HFIAS scores across households with different levels of monthly household income.

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	H₀₂: The regression coefficients for Household Size, Number of Income Earners, and Monthly Household Food Expenditure in predicting HFIAS scores are equal to zero. Methodology This part displays the Research Method Used, Population Frame, Sampling Scheme, Description of the Respondents, Instrument Used, Data Gathering Procedures, and Statistical Treatment of Data. Research Method Used The Researchers employed a cross-sectional design aimed at establishing whether a difference in the level of household food insecurity exists across distinct categories of monthly household income. This approach allowed for the comparison of group differences while statistically managing the influence of other potentially confounding variables, known as covariates. The analysis implemented Quade’s Rank-Based Nonparametric ANCOVA with N = 50, an alternative given that the dataset demonstrated substantial deviations from the assumptions of normality, linearity, and homoscedasticity required by the traditional ANCOVA model; Quade’s procedure provides a robust alternative by ranking the covariate-adjusted scores and performing an ANOVA on the resulting residuals, ensuring the validity of group comparisons under assumption violations (Graphpad Ir & Abolfazl Ghoo djani, 2023).	

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	Population Frame and Sampling Scheme The study targeted household respondents residing in Mandaluyong City, especially adults or parents who manage or contribute to household food decisions. The population frame consisted of households located across multiple barangays within the city (Mandaluyong City Profile – PhilAtlas, n.d.). Data collection was carried out through face-to-face administration of standardized questionnaires during afternoon and evening hours, while some respondents were also approached within their households. To ensure that the sample reflected households with varying levels of food security and socioeconomic conditions, the researchers applied a combined sampling approach. Purposive sampling was used by identifying and targeting household heads or primary caregivers who are responsible for household food management. This allowed the researchers to collect responses from individuals with direct knowledge of food purchases, expenses, and experiences of food insecurity. Convenience sampling was employed to select respondents who were readily accessible and willing to participate during the survey period((Creswell & Creswell, 2018). This approach was necessary due to time constraints, limited resources, and differences in household availability across barangays. The total number of respondents was distributed equally among the researchers, with data collection carried out independently while following the same standardized procedure and eligibility criteria.	

Description of the Respondents

A total of 50 respondents from various households in Mandaluyong City participated in the study. Their socioeconomic characteristics are summarized and described below based on four indicators: estimated monthly household income, household size, number of income earners, and monthly household food expenditure. These variables were collected to provide context to the level of food security experienced by each household and to support the ANCOVA analysis by serving as independent variables or covariates.

The respondents in this study provided information regarding their socioeconomic characteristics, beginning with their estimated monthly household income ranging below ₱25,000 (n=25, 50%), ₱25,001-40,000 (n=17, 34%), and above ₱40,001 (n=8, 16%), where they indicated which of the three income brackets their household belongs to. They also identified their household size, specifying whether their family consists of 1–3 (n=18, 36%), members, 4–5 (n=19, 38%), members, or 6 or more members (n=13, 26%). In addition, respondents reported that the number of income earners in their households is 1-3(n= 40, 80%), 4-5(n=10, 20%), stating whether there is only one breadwinner, two earners, or three and above who contribute financially. Lastly, they were asked to provide their monthly household food expenditure categorized as ₱4,999 (n=8, 16%), ₱5,000-₱7,999 (n=13, 26%), ₱8,000-₱11,999 (n=23, 46%), ₱12,000-₱17,999 (n=4, 8%),

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	₱18,000 above (n=2, 4%) estimating how much their family spends on food each month based on the available categories. Instrument Used The instrument used in this study was a self-administered survey questionnaire composed of two major sections: (1) Socioeconomic Profile and (2) Household Food Insecurity Access Scale (HFIAS). The questionnaire was developed by the researchers based on established food security measurement tools and adapted to the context of households in Mandaluyong City. The HFIAS, originally developed by Coates, Swindale & Bilinsky (2007), continues to be widely used in recent studies (e.g., Angeles-Agdeppa et al., 2021; Protacio et al., 2025; Gebreyesus et al., 2015), including in the Philippines, validating its continued relevance and applicability for measuring household food insecurity. 1. Socioeconomic profile This section gathered essential background information about each respondent's household. The variables included estimated monthly household income, household size, Number of income earners, and monthly household food expenditure. 2. Household Food Insecurity Access Scale (HFIAS) All responses are ordinal. This section evaluates respondents' food insecurity. It seeks to identify the population of food-insecure people and assesses	

their impact on the studies. It asked how often the respondents experience food insecurity with the scale and range assessing.

Scale	Interpretation
0	No
1	Rarely
2	Sometimes
3	Often

Data Gathering Procedures

The data for this study were collected using a structured questionnaire composed of two sections: (1) Socioeconomic Profile and (2) the Household Food Insecurity Access Scale (HFIAS). The data-gathering procedures followed several sequential phases to ensure accuracy, ethical compliance, and statistical appropriateness.

The preliminary phase involved the development, validation, and pilot administration of the survey instrument. The questionnaire items were adapted from standardized tools, primarily the HFIAS by Coates, Swindale & Bilinsky (2007). A Filipino translation was prepared to ensure clarity for respondents with limited literacy. The finalized questionnaire was administered to households within the target area. Data collection was conducted face-to-face to ensure complete responses and to help respondents who may have difficulty reading or interpreting items.

Data Processing and Analysis. Collected responses were encoded into statistical software (SPSS v.28) for analysis. Data cleaning was performed to ensure accuracy, removing incomplete or inconsistent responses. The final dataset was subjected to statistical analysis to draw meaningful conclusions regarding digital food platform adoption and its impact on food security.

Statistical Treatment of Data

Frequencies and percentages: The distribution of categorical variables, such as Estimated Monthly Household Income, as well as the classifications for Household Size and Number of Income Earners, were summarized using frequencies (n) and percentages (%).

Medians (instead of means) for the dependent variable (DV) across groups, to measure central tendency—the raw mean of the original dependent variable is not core descriptive statistics for Quade's Nonparametric Ancova, medians are more robust measures of central tendency, which may occasionally see that mean ranks reported in the output (Edward, 2019), these are the averages of the ranked values of the DV (or covariate).

$$\text{Median} = L + \left(\frac{\frac{N}{2} - F}{f} \right) \times w$$

Where:

- L: Lower boundary of the median class

- N: Total Number of observations (Sample Size)
- F: Cumulative frequency of the class preceding the median class
- f: Frequency of the median class
- w: Width of the median class

Interquartile ranges (IQR) (Instead of standard deviations) for the DV to measure spread. It is the difference between the 75th percentile (Q3) and the 25th percentile (Q1). It is the robust measure of variability preferred alongside the median for nonparametric data.

Where: $IQR = Q3 - Q1$

QUADE'S TEST STATISTICS (Quade's Nonparametric ANCOVA): This was the primary test used to determine the effect of the Estimated Monthly Household Income on the HFIAS Score, adjusted for the covariates (Theodorsson-Norheim, 1987).

Quade Scores (Adjusted Ranks)

$$F = \frac{MS_{Between}}{MS_{Within}}$$

Where:

F: The F-statistic value.

MSBetween: Mean Square (Variance) between the three Income Groups.

MSwithin: Mean Square (Variance) within the groups (the error term).

Results and Discussion

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The data is presented and analyzed in tabular format. The results from the statistical calculations were interpreted using quantitative analysis and organized systematically in relation to the Statements of the Problem.

Table 1: Frequency and Percentage Distribution of the Respondents according to Estimated Monthly Household Income

Estimated Monthly Household Income	(n)Frequency	(%)Percentage
Below ₱25,000	25	50%
₱25,001 – ₱40,000	17	34%
₱40,001 and above	8	16%

Half of the respondents (50%) fall under the below ₱25,000 income category, indicating a predominantly low-income sample. Meanwhile, 34% earn ₱25,001–₱40,000, and only 16% are in the highest income bracket (₱40,001+), reflecting limited representation from higher-income households.

Table 2: Frequency and Percentage Distribution of the Respondents according to household size

Household Size	(n)Frequency	(%)Percentage
1-3	18	36%
4-5	19	38%
6+	13	26%

Half of the households (50%) have six or more members, making large families the most common. Meanwhile, 26% have 1–3 members and 24% have 4–5 members.

Table 3: Frequency and Percentage Distribution of the Respondents according to Number of Income Earners

Number of Income earners	(n)Frequency	(%)Percentage
1-3	40	80%
4-5	10	20%
6+	0%	0%

A large majority of respondents (80%) reported having 1–3 income earners, whereas only 20% have 4–5 earners. This suggests that most households depend on a relatively small earning group.

Table 4: Frequency and Percentage Distribution of the respondents according to Monthly Household Food Expenditure

Monthly Household Food Expenditure	(n) Frequency	(%) Percentage
Below ₱4,999.00	8	16%
₱5,000.00-₱7,999.00	13	26%
₱ 8,000.00-₱11,999.00	23	46%
₱12,000.00-₱17,999.00	4	8%
Above ₱18,000	2	4%

The majority of respondents (46%) fall under moderate food expenditure (₱8,000–₱11,999), consistent with the typical national household food budget. Lower expenditure brackets (very low and low) account for a combined 42%, while only 12% report higher food spending, suggesting that most households operate within limited food budgets.

Table 5: The significant difference in HFIAS scores across households with different levels of monthly household income by Univariate Analysis: Test of Between-Subjects effects, Post-Hoc Test, and Homogeneous Subsets

Tests of Between-Subjects Effects

Dependent Variable: Unstandardized Residual

Source	Type III Sum of Squares	df	Mean Square	F	Sig.
Corrected Model	1675.140 ^a	2	837.570	5.839	.005
Intercept	159.066	1	159.066	1.109	.298
EstimatedMonthlyHouseholdIncome	1675.140	2	837.570	5.839	.005
Error	6741.695	47	143.440		
Total	8416.835	50			
Corrected Total	8416.835	49			

a. R Squared = .199 (Adjusted R Squared = .165)

Based on the evaluation of the estimated monthly household income effect on the dependent variable after adjusting for the covariate, *there is a statistically significant effect of income level on the adjusted dependent variable*, $F(2, 47) = 5.84$, $p = .005$, suggesting that participants' scores differed significantly across income groups even after controlling for the covariate. The model explained approximately 20% of the variance in the outcome ($R^2 = .199$)

Multiple Comparisons

Dependent Variable: Unstandardized Residual

Tukey HSD

(I) EstimatedMonthlyHouseh oldIncome	(J) EstimatedMonthlyHouseh oldIncome	Mean Difference (I- J)	Std. Error	Sig.	95% Confidence Interval	
					Lower Bound	Upper Bound
1	2	11.4106699 [*]	3.76500711	.011	2.2988948	20.5224451
	3	11.9134745 [*]	4.86494117	.047	.1397252	23.6872238
2	1	-11.4106699 [*]	3.76500711	.011	-20.5224451	-2.2988948
	3	.5028046	5.13494942	.995	-11.9243976	12.9300067
3	1	-11.9134745 [*]	4.86494117	.047	-23.6872238	-1.1397252
	2	-.5028046	5.13494942	.995	-12.9300067	11.9243976

Based on observed means.

The error term is Mean Square(Error) = 143.440.

*. The mean difference is significant at the 0.05 level.

Going further, Tukey-adjusted post-hoc comparisons revealed that the lowest income group (Group 1) had significantly higher adjusted residual scores than both Group 2 ($p = .011$) and Group 3 ($p = .047$). No significant difference was found between Groups 2 and 3 ($p = .995$). These results indicate that, after adjustment for the covariate, participants in the lowest income category differed significantly from those in the two higher income categories, while the two higher groups did

not differ
from each
other.

In
more
depth of
analysis,
Post hoc

Unstandardized ResidualTukey HSD^{a, b, c}

EstimatedMonthlyHouseh oldIncome	N	Subset	
		1	2
3	8	-6.1276908	
2	17	-5.6248862	
1	25		5.7857837
Sig.		.994	1.000

Means for groups in homogeneous subsets are displayed.

Based on observed means.

The error term is Mean Square(Error) = 143.440.

a. Uses Harmonic Mean Sample Size = 13.403.

b. The group sizes are unequal. The harmonic mean of the group sizes is used. Type I error levels are not guaranteed.

c. Alpha = 0.05.

analyses using Tukey's HSD on the unstandardized residuals indicated that the lowest-income group (Group 1) had significantly higher adjusted food insecurity scores ($M = 5.79$) than both the middle-income group (Group 2; $M = -5.62$) and the highest-income group (Group 3; $M = -6.13$). The middle- and high-income groups did not significantly differ from each other, as they were placed in the same homogeneous subset. These results suggest that, even after controlling for covariates, households with the lowest income experience significantly greater food insecurity compared to higher-income groups.

Table 6: The covariates' prediction on the HFIAS based on the Correlation, Model Summary, Coefficient, and Residual Statistics

		Correlations			
		Rank of QuestionsTotalScore	Rank of HouseholdSize	Rank of NumberOfIncomeearners	Rank of MonthlyHouseholdFoodExpenditure
Pearson Correlation	Rank of QuestionsTotalScore	1.000	.291	-.124	-.205
	Rank of HouseholdSize	.291	1.000	.201	.157
	Rank of NumberOfIncomeearners	-.124	.201	1.000	.100
	Rank of MonthlyHouseholdFoodExpenditure	-.205	.157	.100	1.000
Sig. (1-tailed)	Rank of QuestionsTotalScore	.	.020	.195	.077
	Rank of HouseholdSize	.020	.	.081	.139
	Rank of NumberOfIncomeearners	.195	.081	.	.245
	Rank of MonthlyHouseholdFoodExpenditure	.077	.139	.245	.
N	Rank of QuestionsTotalScore	50	50	50	50
	Rank of HouseholdSize	50	50	50	50
	Rank of NumberOfIncomeearners	50	50	50	50
	Rank of MonthlyHouseholdFoodExpenditure	50	50	50	50

A Pearson correlation analysis was performed using ranked data. The results showed that household size was significantly and positively correlated with

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the ranked total score, $r(48) = .29$, $p = .020$ (one-tailed). However, neither the number of income earners ($r(48) = -.12$, $p = .195$) nor monthly household food expenditure ($r(48) = -.21$, $p = .077$) showed significant correlations with the ranked total score. No other significant relationships were observed among the predictor variables.

Model Summary^b

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	R Square Change	Change Statistics			Sig. F Change
						F Change	df1	df2	
1	.421 ^a	.177	.123	13.526813	.177	3.299	3	46	.029

a. Predictors: (Constant), Rank of MonthlyHouseholdFoodExpenditure, Rank of NumberOfIncomeearners, Rank of HouseholdSize

b. Dependent Variable: Rank of QuestionsTotalScore

In the Model Summary, it revealed that the ranked covariates—monthly household food expenditure, number of income earners, and household size—significantly predicted the ranked questionnaire scores, $R = .42$, $R^2 = .18$, adjusted $R^2 = .12$, $F(3, 46) = 3.30$, $p = .029$.

ANOVA^a

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	1811.165	3	603.722	3.299	.029 ^b
	Residual	8416.835	46	182.975		
	Total	10228.000	49			

a. Dependent Variable: Rank of QuestionsTotalScore

b. Predictors: (Constant), Rank of MonthlyHouseholdFoodExpenditure, Rank of NumberOfIncomeearners, Rank of HouseholdSize

A multiple regression analysis also showed that the predictors—rank of monthly household food expenditure, rank of number of income earners, and rank of household size—significantly predicted the rank of total question score, $F(3, 46)$

= 3.30, $p = .029$. The model explained 17.7% of the variance in the dependent variable.

Coefficients ^a											
Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	Correlations			Collinearity Statistics	
		B	Std. Error	Beta			Zero-order	Partial	Part	Tolerance	VIF
1	(Constant)	28.594	6.508		4.394	.000					
	Rank of HouseholdSize	.384	.146	.364	2.637	.011	.291	.362	.353	.941	1.063
	Rank of NumberOfIncomeearners	-.247	.196	-.173	-1.263	.213	-.124	-.183	-.169	.955	1.047
	Rank of MonthlyHouseholdFoodExpenditure	-.258	.143	-.244	-1.799	.079	-.205	-.256	-.241	.971	1.030

a. Dependent Variable: Rank of QuestionsTotalScore

The overall model was significant, $F(3, 46) = 3.30$, $p = .029$, explaining 17.7% of the variance. Household size significantly predicted the rank of total question score ($\beta = .36$, $p = .011$), with larger household size ranks associated with higher total score ranks. Neither the number of income earners ($\beta = -.17$, $p = .213$) nor monthly household food expenditure ($\beta = -.24$, $p = .079$) significantly predicted the outcome. Multicollinearity diagnostics indicated no concerns (VIFs ≈ 1).

Residuals Statistics ^a					
	Minimum	Maximum	Mean	Std. Deviation	N
Predicted Value	15.17650	39.25621	25.50000	6.079684	50
Residual	-24.404896	25.823502	.000000	13.106187	50
Std. Predicted Value	-1.698	2.263	.000	1.000	50
Std. Residual	-1.804	1.909	.000	.969	50

a. Dependent Variable: Rank of QuestionsTotalScore

Furthermore, Residual Statistics showed residuals exhibited a mean of 0.00 (SD = 13.11), indicating that the model's predictions were unbiased on average, though residual variability was considerable. The standardized residuals ranged from -1.80 to 1.91, with a mean of 0.00 (SD = 0.97), suggesting that most residuals

fell within two standard deviations of the mean, a common criterion for assessing the quality of a model.

Conclusion and Recommendations

Conclusion

Based on the findings, the following conclusions were drawn.

1. **Most respondents come from low-income, large households with limited earning capacity.** Half earn below ₱25,000, and many have six or more members but only one to three income earners, indicating high dependency and financial strain. Their food spending is generally modest, reflecting constrained household budgets.
2. **Monthly household income significantly affects food insecurity levels.** After controlling for the covariate, the lowest-income households showed significantly higher HFIAS scores than middle- and high-income groups, which did not differ from each other. This confirms that lower income is strongly associated with greater food insecurity.
3. **Household size is the only significant predictor of food insecurity among the covariates.** Larger households exhibit higher HFIAS scores, while the number of income earners and food expenditure do not significantly predict food insecurity. Although the model explained 18% of the variance, the findings emphasize household size as the primary factor influencing food insecurity in this sample.

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	Recommendations Based on the study’s findings, the following recommendations are proposed to address the factors associated with household food insecurity: 1. Since most households are low-income, large, and financially strained, assistance programs should prioritize these vulnerable groups. Efforts may include livelihood and income-enhancing initiatives, food and cash assistance, and financial or nutrition education to help families manage limited budgets more effectively. Programs should also consider household size to ensure sufficient support for larger families with high dependency ratios. 2. Because lower-income households experience significantly greater food insecurity, interventions should be income-sensitive and targeted toward the poorest income groups. Local governments and agencies should strengthen food assistance, nutritional programs, and social protection specifically for families earning the least. Income indicators should also be integrated into food security monitoring systems for better targeting and resource allocation. 3. Given that household size is the only significant predictor of food insecurity, food security programs should allocate resources proportionately based on family size. Larger households should receive enhanced or scaled-up support. Future research should also explore additional variables—such as education, employment stability, and access to services—to improve	

understanding of other factors that may influence food insecurity beyond household size.

Appendices

Appendix I: Survey Instrument

Dear Respondent,

We are conducting a research study titled **"Food Security and Poverty: Assessing Household Experiences of Food Shortage."** This study aims to understand the challenges households face in accessing food and how these experiences relate to their economic conditions.

You have been selected as a respondent because your experiences and background are relevant to the focus of this research. Your participation is greatly valued and will contribute significantly to the accuracy and relevance of our findings.

Please be assured that all your responses will be used strictly for academic purposes and will not be shared with any third party. You may choose to skip any question you are not comfortable answering.

Thank you very much for your time, honesty, and cooperation. Your input plays an important role in the completion of this study.

Sincerely,

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Section 1: Socioeconomic Profile

1. Estimated Monthly Household Income (Tinatayang Buwanang Kita ng Sambahayan):

- (1) ☐ Below ₱25,000 (Mas mababa sa ₱25,000)
(2) ☐ ₱25,001 – ₱40,000
(3) ☐ ₱40,001 and above (₱40,001 pataas)

2. Household Size (Bilang ng Tao sa Bahay):

- (1) ☐ 1-3
(2) ☐ 4-5
(3) ☐ 6+

3. Number of Income Earners (Bilang ng Kumakita sa Sambahayan):

- (1) ☐ 1-3
(2) ☐ 4-5
(3) ☐ 6+

4. Monthly Household Food Expenditure (Buwanang Gastos sa Pagkain):

- (1) ☐ Below ₱4,999.00 (very low food spending)
(2) ☐ ₱5,000.00-₱7,999.00 (low-to-moderate spending)
(3) ☐ ₱8,000.00-₱11,999.00 (around the national mean)
(4) ☐ ₱12,000.00-₱17,999.00 (moderate-to-higher spending)
(5) ☐ Above ₱18,000 (high food spending)

Section 2: Household Food Insecurity Access Scale (HFIAS)

Instructions: For each of the following questions, please indicate whether this situation occurred in your household in the past 30 days, and if so, how often it happened. (Mga Tagubilin: Para sa bawat isa sa mga sumusunod na tanong, mangyaring ipaliwalig kung ang sitwasyong ito ay nangyari sa iyong sambahayan sa nakalipas na 30 araw, at kung gayon, gaano kadalas ito nangyari.)

	No	Rarely	Sometimes	Often
1. Did you worry that your household would not have enough food? (Nag-aalala ka ba na ang iyong sambahayan ay walang sapat na pagkain?)				
2. Were you or any household members unable to eat preferred foods or a varied diet due to lack of resources? (Ikaw ba o sinumang miyembro ng sambahayan ay hindi nakakain ng mga gustong pagkain o iba't ibang diyeta dahil sa kakulangan ng mapagkukunan?)				
3. Did you or any household members eat smaller meals than needed because there was not enough food? (Ikaw ba o sinumang miyembro ng sambahayan ay kumain ng mas maliit na pagkain kaysa sa kinakailangan dahil walang sapat na pagkain?)				
4. Did you or any household members go to sleep hungry because there was not enough food? (Ikaw ba o sinumang miyembro ng sambahayan ay natutulog nang gutom dahil walang sapat na pagkain?)				
5. Was there ever no food in your household because of a lack of resources? (Wala bang pagkain sa iyong sambahayan dahil sa kakulangan ng mga mapagkukunan?)				
6. Did you or any household members go a whole day and night without eating because there was not enough food? (Ikaw ba o sinumang miyembro ng sambahayan ay umaalis sa buong araw at gabi na hindi kumakain dahil walang sapat na pagkain?)				

Appendix 2: Data Output in SPSS

Rank

Created Variables^a

Source Variable	Function	New Variable	Label
QuestionsTotalScore ^b	Rank	RQuestio	Rank of QuestionsTotalScore
HouseholdSize ^b	Rank	RHouseho	Rank of HouseholdSize
NumberOfIncomeearners ^b	Rank	RNumberO	Rank of NumberOfIncomeearners
MonthlyHouseholdFoodExpenditure ^b	Rank	RMonthly	Rank of MonthlyHouseholdFoodExpenditure

a. Mean rank of tied values is used for ties.

b. Ranks are in ascending order.

Regression**Descriptive Statistics**

	Mean	Std. Deviation	N
Rank of QuestionsTotalScore	25.50000	14.447654	50
Rank of HouseholdSize	25.50000	13.684491	50
Rank of NumberOfIncomeearners	25.50000	10.101525	50
Rank of MonthlyHouseholdFoodExpenditure	25.50000	13.680763	50

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Correlations

		Rank of QuestionsTot alScore	Rank of HouseholdSiz e	Rank of NumberOfInc omeearners	Rank of MonthlyHous eholdFoodEx penditure
Pearson Correlation	Rank of QuestionsTotalScore	1.000	.291	-.124	-.205
	Rank of HouseholdSize	.291	1.000	.201	.157
	Rank of NumberOfIncomeearners	-.124	.201	1.000	.100
	Rank of MonthlyHouseholdFoodE xpenditure	-.205	.157	.100	1.000
Sig. (1-tailed)	Rank of QuestionsTotalScore	.	.020	.195	.077
	Rank of HouseholdSize	.020	.	.081	.139
	Rank of NumberOfIncomeearners	.195	.081	.	.245
	Rank of MonthlyHouseholdFoodE xpenditure	.077	.139	.245	.
N	Rank of QuestionsTotalScore	50	50	50	50
	Rank of HouseholdSize	50	50	50	50
	Rank of NumberOfIncomeearners	50	50	50	50
	Rank of MonthlyHouseholdFoodE xpenditure	50	50	50	50

Variables Entered/Removed^a

Model	Variables Entered	Variables Removed	Method
1	Rank of MonthlyHous eholdFoodEx penditure, Rank of NumberOfInc omeearners, Rank of HouseholdSiz e ^b	.	Enter

a. Dependent Variable: Rank of
QuestionsTotalScore

b. All requested variables entered.

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Model Summary^b

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	R Square Change	Change Statistics			Sig. F Change
						F Change	df1	df2	
1	.421 ^a	.177	.123	13.526813	.177	3.299	3	46	.029

a. Predictors: (Constant), Rank of MonthlyHouseholdFoodExpenditure, Rank of NumberOfIncomeearners, Rank of HouseholdSize

b. Dependent Variable: Rank of QuestionsTotalScore

ANOVA^a

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	1811.165	3	603.722	3.299	.029 ^b
	Residual	8416.835	46	182.975		
	Total	10228.000	49			

a. Dependent Variable: Rank of QuestionsTotalScore

b. Predictors: (Constant), Rank of MonthlyHouseholdFoodExpenditure, Rank of NumberOfIncomeearners, Rank of HouseholdSize

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	Correlations			Collinearity Statistics	
		B	Std. Error	Beta			Zero-order	Partial	Part	Tolerance	VIF
1	(Constant)	28.594	6.508		4.394	.000					
	Rank of HouseholdSize	.384	.146	.364	2.637	.011	.291	.362	.353	.941	1.063
	Rank of NumberOfIncomeearners	-.247	.196	-.173	-1.263	.213	-.124	-.183	-.169	.955	1.047
	Rank of MonthlyHouseholdFoodExpenditure	-.258	.143	-.244	-1.799	.079	-.205	-.256	-.241	.971	1.030

a. Dependent Variable: Rank of QuestionsTotalScore

Collinearity Diagnostics^a

Model	Dimension	Eigenvalue	Condition Index	(Constant)	Variance Proportions		
					Rank of HouseholdSize	Rank of NumberOfIncomeearners	Rank of MonthlyHouseholdFoodExpenditure
1	1	3.605	1.000	.01	.01	.01	.01
	2	.188	4.383	.00	.39	.02	.70
	3	.147	4.954	.04	.55	.38	.16
	4	.060	7.720	.95	.05	.59	.13

a. Dependent Variable: Rank of QuestionsTotalScore

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Residuals Statistics^a

	Minimum	Maximum	Mean	Std. Deviation	N
Predicted Value	15.17650	39.25621	25.50000	6.079684	50
Residual	-24.404896	25.823502	.000000	13.106187	50
Std. Predicted Value	-1.698	2.263	.000	1.000	50
Std. Residual	-1.804	1.909	.000	.969	50

a. Dependent Variable: Rank of QuestionsTotalScore

Univariate Analysis

Between-Subjects Factors

	N	
EstimatedMonthlyHouseholdIncome	1	25
	2	17
	3	8

Tests of Between-Subjects Effects

Dependent Variable: Unstandardized Residual

Source	Type III Sum of Squares	df	Mean Square	F	Sig.
Corrected Model	1675.140 ^a	2	837.570	5.839	.005
Intercept	159.066	1	159.066	1.109	.298
EstimatedMonthlyHouseholdIncome	1675.140	2	837.570	5.839	.005
Error	6741.695	47	143.440		
Total	8416.835	50			
Corrected Total	8416.835	49			

a. R Squared = .199 (Adjusted R Squared = .165)

Post-Hoc Test: Estimated Household Monthly Income

Multiple Comparisons

Dependent Variable: Unstandardized Residual

Tukey HSD

(I) EstimatedMonthlyHouseh oldIncome	(J) EstimatedMonthlyHouseh oldIncome	Mean Difference (I- J)	Std. Error	Sig.	95% Confidence Interval	
					Lower Bound	Upper Bound
1	2	11.4106699*	3.76500711	.011	2.2988948	20.5224451
	3	11.9134745*	4.86494117	.047	.1397252	23.6872238
2	1	-11.4106699*	3.76500711	.011	-20.5224451	-2.2988948
	3	.5028046	5.13494942	.995	-11.9243976	12.9300067
3	1	-11.9134745*	4.86494117	.047	-23.6872238	-1.1397252
	2	-.5028046	5.13494942	.995	-12.9300067	11.9243976

Based on observed means.

The error term is Mean Square(Error) = 143.440.

*. The mean difference is significant at the 0.05 level.

HOMOGENOUS SUBSETS

Unstandardized Residual

Tukey HSD^{a,b,c}

EstimatedMonthlyHouseh oldIncome	N	Subset	
		1	2
3	8	-6.1276908	
2	17	-5.6248862	
1	25		5.7857837
Sig.		.994	1.000

Means for groups in homogeneous subsets are displayed.

Based on observed means.

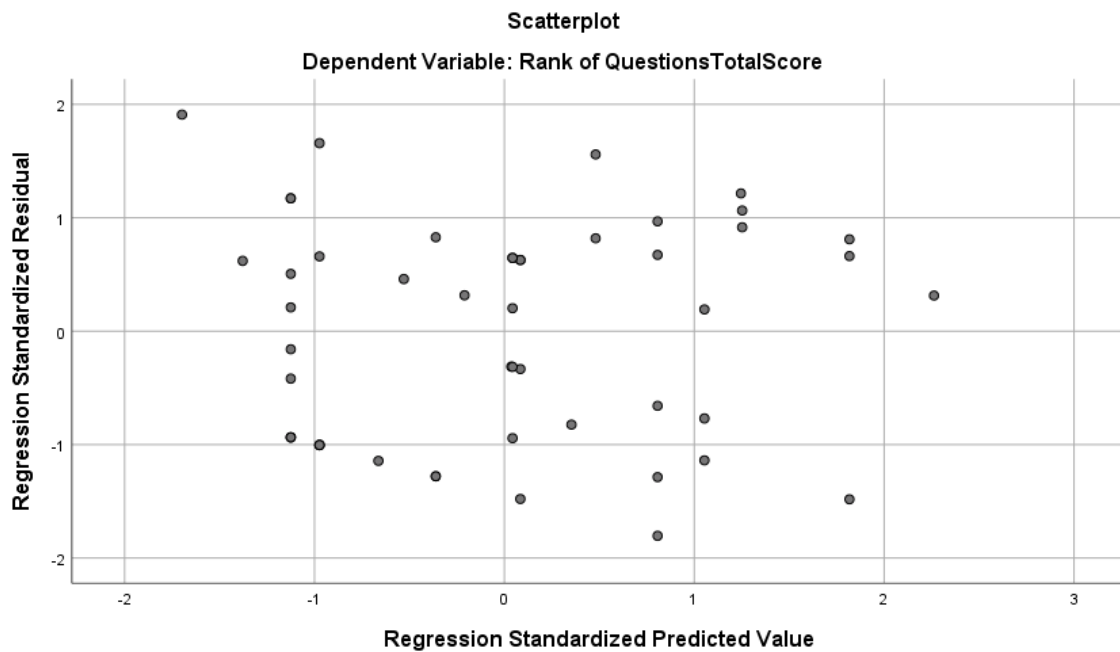
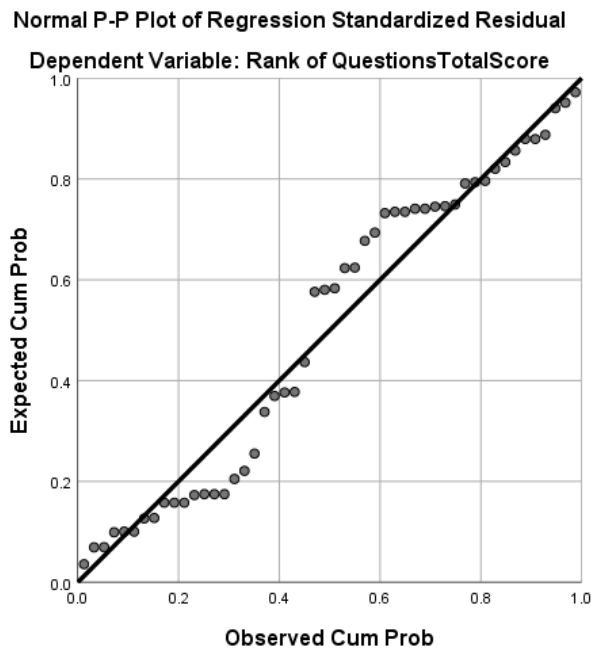
The error term is Mean Square(Error) = 143.440.

a. Uses Harmonic Mean Sample Size = 13.403.

b. The group sizes are unequal. The harmonic mean of the group sizes is used. Type I error levels are not guaranteed.

c. Alpha = 0.05.

Appendix C: Charts



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